Showing our fibre network on a web map: 3 options compared

We need a web map that shows our fibre sites and the cable routes between them. This page compares three ways to build it, using the same mock data (5 sites and 4 routes around Kajang, Cheras, KL, Bangi and Putrajaya).

On every map, routes are coloured by type: ▬ Underground (brown, solid) and ▬ Aerial (blue, dashed). Click any site or route to see its name and details.

Printed copy: the live maps are left out. See the Screenshots section.

Screenshots

Captured with Playwright (headless Chromium) at 1400 × 800 on 15 Oct 2026, so this record still works when the servers are off. Click an image for full size.

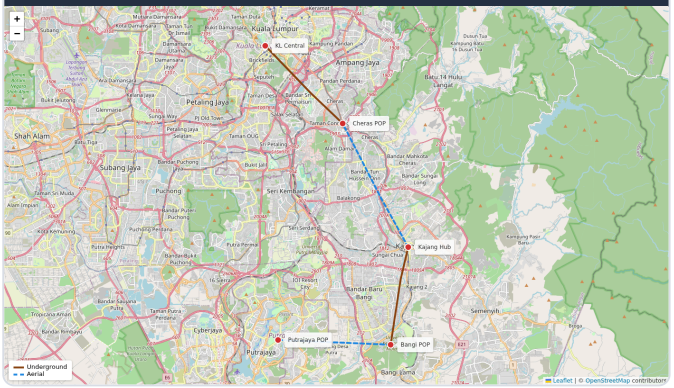
Whole network

Google Maps

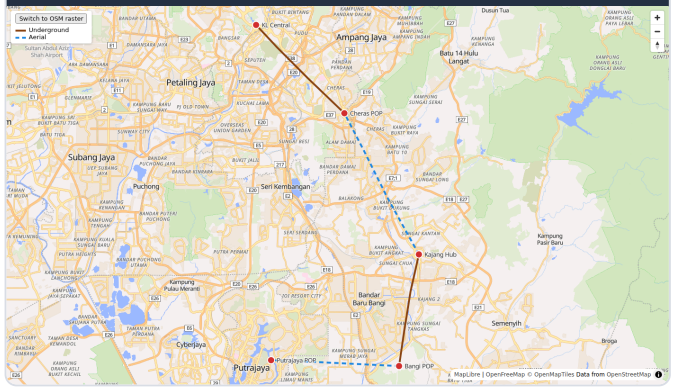
Not tested

Needs a Google API key and a billing account with a card.

Leaflet + OpenStreetMap



MapLibre + OpenFreeMap



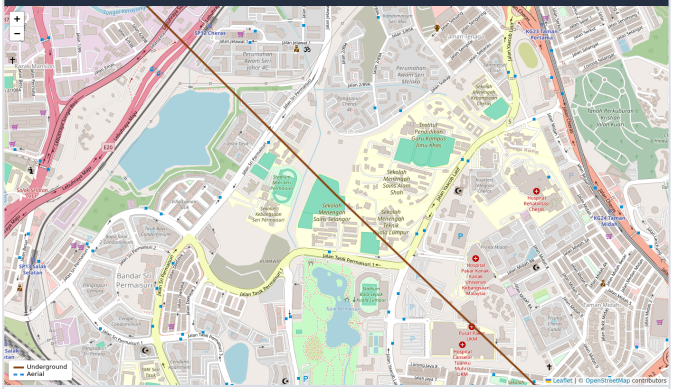
Close-up on one route (Cheras to KL Central, same scale)

Google Maps

Not tested

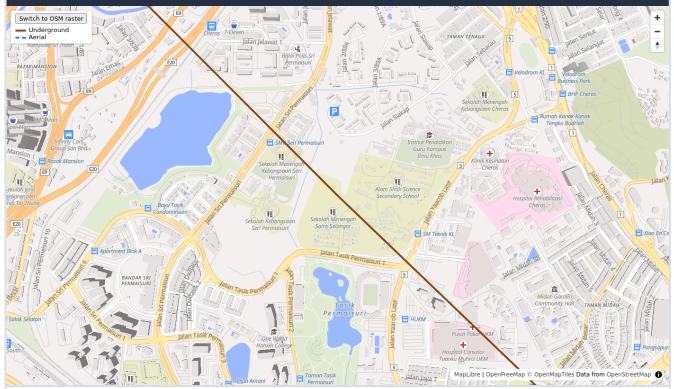
Needs a Google API key and a billing account with a card.

Leaflet + OpenStreetMap



Raster images: text and lines are fixed pixels and blur when scaled between zoom levels.

MapLibre + OpenFreeMap



Vector data drawn by the browser: text and lines stay sharp at any zoom.

Comparison

	Google Maps	Leaflet + OSM tiles	MapLibre + OpenFreeMap
Look	Most polished and familiar basemap	Flat raster tiles, looks dated, blurs between zoom levels	Smooth vector rendering, crisp at any zoom, rotation and tilt
Route styling	Data layer only: colour and width, no dashes	Full control, dashes included	Full control, data-driven styling, dashes via separate layers
Cost	Free monthly allowance, then pay per use	Free	Free
Key or account needed	API key plus billing account with a card	None	None
Integration effort	Script tag plus key, loadGeoJson	Easiest, about 20 lines	Slightly more: style, sources and layers

	Google Maps	Leaflet + OSM tiles	MapLibre + OpenFreeMap
Scales to many routes	Fine for hundreds	Fine for hundreds to low thousands, slows after that	Best: WebGL handles tens of thousands
Risks	Vendor lock-in, key management, terms of use	OSM's public tile server forbids heavy use	OpenFreeMap has no SLA unless you self-host
Self-hosting/offline	No	Yes, with your own tiles	Yes, with your own tiles and style
PostGIS path	GeoJSON	GeoJSON	GeoJSON now, vector tiles later (Martin or pg_tileserv)